

Project Context: POD Customization Engine

Intelligent Personalization & Print-on-Demand Platform

1. System Vision

The core objective is to enable users to design and customize products using **Generative AI** and **Direct Uploads**. The system follows an "AI-first" philosophy where the AI acts as a creative assistant, not just a feature.

2. Technical Architecture Stack

Framework: Antigravity (Python)

Frontend: Next.js + Tailwind CSS

Backend: FastAPI (Python)

Database: Supabase (PostgreSQL)

Storage: Supabase Storage (S3-compatible)

Canvas: Fabric.js / Konva.js

3. Core Data Entities (per UML)

Entity	Key Responsibilities
Client	Handles product selection, design customization, and order tracking.
Design	Stores file URLs, prompt history, format types, and resolution metadata.
Commande	Manages the lifecycle of an order, including total price and status.
LigneCommande	The "bridge" storing specific customization coordinates (x, y, scale).

4. Security & Compliance (Cybersecurity Focus)

- **Input Validation:** Strict sanitization of AI prompts to prevent injection.
- **File Security:** Magic number verification for image uploads to prevent malicious payloads.

- **Access Control:** Row Level Security (RLS) via Supabase for design ownership.